

Nucleotide Chemistry & Metabolism

WBUHS BIOCHEMISTRY · PAPER 1

Purine & Pyrimidine Structure

De Novo Synthesis

Salvage Pathway

Uric Acid & Gout

Allopurinol

Lesch-Nyhan Syndrome

ADA Deficiency

Nucleotide Drugs

CPS I vs CPS II

Orotic Aciduria

01 Nucleotide Structure

⚡ A **Nucleotide** = Nitrogenous base + Pentose sugar + Phosphate group.
Remove phosphate →

Nucleoside . Remove sugar → just the base.

Purines

Bicyclic: pyrimidine fused to imidazole ring

- ♦ **Adenine (A)** — amino group at C6
- ♦ **Guanine (G)** — keto group at C6, amino at C2
- ♦ Found in both DNA and RNA
- ♦ Larger molecules (2 rings)

Pyrimidines

Monocyclic: single 6-membered ring

- ♦ **Cytosine (C)** — both DNA & RNA
- ♦ **Thymine (T)** — DNA only (methyl group at C5)
- ♦ **Uracil (U)** — RNA only (no methyl group)
- ♦ Smaller molecules (1 ring)

Sugars

Pentose in N-glycosidic bond (β -configuration)

- ♦ **Ribose** — in RNA nucleotides (2'-OH present)
- ♦ **Deoxyribose** — in DNA (2'-H, no OH)
- ♦ Base attached to C1' (N-glycosidic bond)
- ♦ Phosphate on 5'-OH

Key Bonds

Connecting components

- ♦ **N-glycosidic bond:** base → sugar (C1')
- ♦ **Phosphoester bond:** sugar C5' → phosphate
- ♦ **3'-5' phosphodiester:** links nucleotides in chain
- ♦ DNA more stable: 2'-H (not OH) → alkali resistant

COMPONENT	PURINE	PYRIMIDINE
Ring structure	Bicyclic (double ring)	Monocyclic (single ring)
Members	Adenine, Guanine	Cytosine, Thymine (DNA), Uracil (RNA)
Pairing (Watson-Crick)	A pairs with T (2 H-bonds); G pairs with C (3 H-bonds)	Paired with purines
De novo synthesis	On pre-formed ribose-5-P scaffold (IMP → AMP/GMP)	Ring built first, then attached to ribose
End product of catabolism	Uric acid (humans)	CO₂ + NH₃ + β-alanine/β-aminoisobutyrate
Salvage pathway enzyme	HGPRT (HG), APRT (A)	Not prominent (minor UPRTase)

🟡 MNEMONIC – WHICH BASES IN DNA VS RNA

DNA: "A TiGht Cell" = A, T, G, C (Thymine only in DNA)

RNA: "A UGly Cell" = A, U, G, C (Uracil replaces Thymine)

Key EQ: "DNA is alkali resistant, RNA is alkali labile" — because RNA has 2'-OH which forms cyclic phosphodiester intermediate under alkaline hydrolysis; DNA lacks 2'-OH so cannot.

SOURCES OF ATOMS IN THE PURINE RING

★ PYQ 2013, 2015-S

► PURINE RING ATOM ORIGINS (MUST MEMORISE)

Nitrogen atoms (N1, N3, N9)

N1 ← Aspartate

N3, N9 ← Glutamine

N7 ← Glycine

Carbon atoms

C4, C5, N7 ← **Glycine** (entire unit)C2, C8 ← N¹⁰-formyl-THFC6 ← CO₂

💡 MNEMONIC – PURINE RING SOURCES

"Glutamine Attacks Good Cells, Aspartate Fights CO₂"**G**lycine (C4,C5,N7) · **G**lutamine (N3,N9 + amide N of C2,C6) · N¹⁰-Formyl-THF (C2,C8) · **A**spartate (N1) · CO₂ (C6)

► PURINE DE NOVO SYNTHESIS – KEY STEPS TO IMP

Starting material (Step 1)

Ribose-5-phosphate (from PPP)

Activated by PRPP synthetase → PRPP (5-phosphoribosyl-1-pyrophosphate)

↓ **Glutamine:PRPP amidotransferase [RATE LIMITING STEP ⚡]****5-phosphoribosylamine (PRA)**

First committed step · Inhibited by AMP, GMP, IMP (feedback inhibition)

↓ **9 more enzymatic steps (add Glycine, Formyl group, close ring, etc.)**

Intermediate product (10 steps total)

IMP (Inosine Monophosphate) — the common intermediate

Hypoxanthine base + ribose-5-P · Bifurcation point

↓ **Branching**IMP → Adenylosuccinate → **AMP**

Requires GTP + Aspartate · Adenylosuccinate synthetase + lyase

IMP → Xanthosine-5-P → **GMP**

Requires ATP + Glutamine · IMP dehydrogenase + GMP synthetase

🔑 REGULATION OF PURINE DE NOVO SYNTHESIS

- ◆ **PRPP synthetase:** inhibited by AMP, GMP, IMP (end-product inhibition)
- ◆ **Glutamine:PRPP amidotransferase (RLS):** inhibited by AMP + GMP together (synergistic feedback)
- ◆ **Cross-regulation:** AMP synthesis needs GTP; GMP synthesis needs ATP → reciprocal balance
- ◆ In **Gout:** hyperactive PRPP synthetase or mutant HGPRT → excess purine → excess uric acid

⚡ Site of de novo synthesis: **Liver** primarily (can export nucleotides to other tissues). Brain and muscle have limited capacity → depend on salvage.

03 Pyrimidine De Novo Synthesis

⚡ Key difference from purine synthesis: In pyrimidine synthesis, the **ring is built FIRST**, then attached to PRPP. In purines, the ring is built ON the sugar.

ATOM SOURCES IN PYRIMIDINE RING

N1
Aspartate

C2
CO₂ (from HCO₃⁻)

N3
Glutamine

C4, C5, C6
Aspartate

🗨 QUICK RECALL

Pyrimidine = "Aspartate + CO₂ + Glutamine" ("ACG does the ring")

Compare purines (7 donors including glycine, THF). Pyrimidines simpler — only 3 donors.

▶ PYRIMIDINE DE NOVO — KEY STEPS

Step 1 — CPS II [RATE LIMITING ⚡]

Glutamine + CO₂ + 2ATP → **Carbamoyl phosphate**

In cytosol · CPS II (not CPS I which is in mitochondria for urea cycle)

↓ + Aspartate (ATCase / Aspartate transcarbamoylase)

Carbamoyl aspartate → ring closure → **Dihydroorotate**

In cytosol (mammalian cells) — trifunctional CAD enzyme (CPS II + ATCase + DHOase)

↓ Dihydroorotate dehydrogenase (DHODH) — in inner mitochondrial membrane

Orotate

First pyrimidine with aromatic ring

↓ + PRPP (UMP synthase = OPRT + ODCase — bifunctional)

UMP (Uridine Monophosphate) — common intermediate

UMP → UDP → UTP → CTP (by CTP synthetase)

↓ Thymidylate synthase + DHFR pathway

TMP (dTMP) → dTDP → dTTP

dUMP + N⁵,N¹⁰-methylene-THF → dTMP (requires folate as C donor)

🔗 CLINICAL LINK — OROTIC ACIDURIA

Type I (classic): Deficiency of UMP synthase (OPRT + ODCase). Orotate accumulates → excreted in urine. Megaloblastic anaemia (pyrimidines needed for DNA). Treatment: **Uridine** supplementation (bypasses block, also provides feedback inhibition of CPS II).

Type II: Deficiency of ODCase only (less severe).

Distinguish from: Orotic aciduria in urea cycle disorders (OTC deficiency) — excess carbamoyl phosphate overflows into pyrimidine pathway → orotate ↑. But NO megaloblastic anaemia (UMP synthase intact, can process the orotate).

CPS I VS CPS II

★ ADVANCE TOPIC — OFTEN ASKED

FEATURE	CPS I	CPS II
Location	Mitochondria	Cytosol
Pathway	Urea cycle	Pyrimidine synthesis
Nitrogen donor	NH ₃ (free ammonia)	Glutamine (amide N)
Allosteric activator	N-acetylglutamate (essential)	PRPP
Inhibitor	None major (NAG controls)	UMP (end-product)
ATP used	2 ATP	2 ATP
Clinical deficiency	CPS I deficiency → hyperammonaemia	Part of CAD trifunctional enzyme

⚠️ **Classic exam trap:** OTC deficiency (urea cycle) → excess carbamoyl phosphate leaks from mitochondria into cytosol → CPS II pathway → orotic aciduria WITHOUT megaloblastic anaemia. In orotic aciduria (Type I), you get anaemia.

04

Salvage Pathways

⚡ Salvage = Recycle pre-formed bases or nucleosides back to nucleotides.

Energetically cheaper

than de novo. Crucial in tissues with

limited de novo capacity (brain, RBCs).

► PURINE SALVAGE REACTIONS

Free bases (Hypoxanthine, Guanine, Adenine) from turnover or diet



Enzyme: HGPRT

Hypoxanthine-Guanine Phosphoribosyl TransferaseHypoxanthine + PRPP → **IMP**Guanine + PRPP → **GMP**

Enzyme: APRT

Adenine Phosphoribosyl TransferaseAdenine + PRPP → **AMP****Why Salvage Matters:**

- ✦ Brain relies almost exclusively on salvage (low de novo)
- ✦ Less ATP consumed than de novo synthesis
- ✦ Also recovers bases from nucleic acid degradation
- ✦ Maintains nucleotide pool balance

When Salvage Fails:

- ✦ **HGPRT deficiency** → Lesch-Nyhan Syndrome
- ✦ Hypoxanthine/Guanine cannot be recycled → converted to uric acid instead → hyperuricaemia
- ✦ Also: PRPP accumulates → drives de novo synthesis faster → more purines → more uric acid
- ✦ Brain cannot compensate → severe neurological features

ADENOSINE DEAMINASE (ADA) – CRITICAL ENZYME

Adenosine

↓ **ADA (Adenosine Deaminase)**

Inosine

↓ **Purine Nucleoside Phosphorylase**

Hypoxanthine → (salvage or → Xanthine → Uric Acid)

🔗 **ADA DEFICIENCY → SCID (SEVERE COMBINED IMMUNODEFICIENCY)**ADA absent → dATP accumulates (deoxyadenosine → dAMP → dADP → **dATP**).dATP: **inhibits ribonucleotide reductase** → blocks ALL dNTP synthesis → DNA synthesis halts in *rapidly dividing lymphocytes (T & B cells)* → lymphocyte death → profound immunodeficiency.

Treatment: Enzyme replacement (PEG-ADA), gene therapy (first successful gene therapy target).

05 Purine Catabolism & Uric Acid

► PURINE CATABOLISM – AMP & GMP → URIC ACID

AMP

↓ AMP deaminase
or Adenosine → Inosine (ADA)

GMP

↓ 5'-nucleotidase
→ Guanosine → Guanine

↓ Both converge on Hypoxanthine/Xanthine pathway

IMP → Inosine → Hypoxanthine

AMP → AMP deaminase → IMP (or via Adenosine → Inosine by ADA)

↓ **Xanthine Oxidase** (uses O₂, generates H₂O₂) [KEY ENZYME ⚡]

Xanthine

Hypoxanthine → Xanthine (Xanthine oxidase)
Guanine → Xanthine (Guanase/deaminase)

↓ **Xanthine Oxidase** (again)

URIC ACID (end product in humans)

Normal serum: Males <7 mg/dL · Females <6 mg/dL · Excreted by kidney (2/3) + gut (1/3)

WHY URIC ACID (NOT ALLANTOIN) IN HUMANS?

Most mammals have **uricase** → converts uric acid → allantoin (more soluble). Humans and great apes **lost uricase** during evolution (gene silenced). Result: uric acid is the final product and can crystallise in tissues.

ORGANISM	FINAL PURINE CATABOLIC PRODUCT
Humans, apes, birds	Uric acid (no uricase)
Most mammals	Allantoin (uricase present)
Marine invertebrates	NH ₃ (ammonotelic)

06 Gout & Hyperuricaemia

⚡ Gout = **monosodium urate (MSU) crystals** in joints (especially 1st metatarsophalangeal joint) + soft tissues (tophi) due to sustained hyperuricaemia.

↑ Uric Acid Production:

- ◆ **HGPRT deficiency** (Lesch-Nyhan)
- ◆ **PRPP synthetase overactivity**
- ◆ Increased cell turnover (leukaemia, lymphoma, psoriasis)
- ◆ Von Gierke's disease (G6P → PRPP overproduction)
- ◆ High purine diet (red meat, organ meats, shellfish, beer)
- ◆ Alcohol (lactate competes with urate for renal excretion + increased purine catabolism)
- ◆ Tumour lysis syndrome (chemo)

↓ Uric Acid Excretion:

- ◆ **Renal insufficiency**
- ◆ Low-dose aspirin (competes with urate at proximal tubule)
- ◆ Thiazide diuretics
- ◆ Lactic acidosis (lactate competes at renal tubule)
- ◆ Cyclosporine

🔗 ALCOHOL & GOUT

Alcohol → lactate (NADH ↑ in liver → pyruvate → lactate) → lactate competes with urate for renal tubular secretion → ↓ urate excretion → ↑ serum urate → precipitates gout attack.

Also: Beer has guanosine (purine precursor).

▶ ALLOPURINOL = SUICIDE INHIBITOR OF XANTHINE OXIDASE

Allopurinol (structural analogue of Hypoxanthine)
C7-N8 atoms transposed compared to hypoxanthine

↓ Xanthine oxidase oxidises allopurinol

Oxipurinol (Alloxanthine) — tight-binding product
Binds and IRREVERSIBLY inactivates xanthine oxidase → suicide inhibition

↓ Result

↓ Xanthine oxidase activity → ↓ uric acid formation
Hypoxanthine & xanthine accumulate (more soluble than uric acid) → excreted safely
Also: ↓ PRPP (via feedback) → ↓ de novo purine synthesis

⚠ **Drug interaction alert:** Allopurinol inhibits xanthine oxidase → 6-Mercaptopurine (6-MP) and Azathioprine are normally inactivated by XO. If allopurinol + 6-MP → 6-MP toxicity → dose must be reduced to 25% of normal.

DRUG	MECHANISM	USE
Allopurinol	Suicide inhibitor of XO	Gout (chronic), hyperuricaemia
Febuxostat	Non-purine, selective XO inhibitor	Gout (alternative if allopurinol intolerance)
Probenecid	Blocks urate reabsorption at renal tubule (uricosuric)	Gout (increases excretion)
Colchicine	Inhibits neutrophil tubulin polymerisation → ↓ crystal phagocytosis	Acute gout attack
Rasburicase	Recombinant uricase → allantoin	Tumour lysis syndrome

Von Gierke's disease is associated with hyperuricaemia.

PYQ 2012, 2017

Standard Exam Answer Draft:

- Primary Defect:** Von Gierke's disease (Type Ia Glycogen Storage Disease) is caused by a deficiency of **Glucose-6-phosphatase**, blocking the final step of both glycogenolysis and gluconeogenesis.
- G6P Accumulation:** Glucose-6-phosphate (G6P) accumulates and is shunted into the **Pentose Phosphate Pathway**, generating excess Ribose-5-phosphate → elevated **PRPP** (phosphoribosyl pyrophosphate).
- De Novo Purine Surge:** PRPP is the committed substrate for de novo purine synthesis. Its excess **drives accelerated purine production**, greatly increasing flux through the catabolic pathway.
- Uric Acid Overproduction:** The purine catabolism pathway terminates in **uric acid** (via xanthine oxidase), causing hyperuricaemia.
- Renal Competition:** Lactic acidosis (from blocked gluconeogenesis) causes lactate to compete with urate for renal tubular secretion, further **reducing uric acid excretion** and worsening hyperuricaemia.
- Clinical Result:** Gout and/or uric acid nephrolithiasis develop alongside the typical features of hypoglycaemia and hepatomegaly.

- ◆ **Deficiency:** HGPRT (complete absence)
- ◆ **Inheritance:** X-linked recessive (affects males)
- ◆ **Gene:** HPRT1 gene on X chromosome

Biochemical Consequences:

- ◆ Hypoxanthine & Guanine cannot be salvaged
- ◆ ↑↑ PRPP (no consumption by HGPRT) → drives de novo synthesis
- ◆ ↑↑ Purine catabolism → uric acid

Clinical Features (the triad):

- ◆ Severe **gout** + hyperuricaemia (urate nephropathy, tophi)
- ◆ **Neurological:** intellectual disability, choreoathetosis, spastic cerebral palsy
- ◆ **Compulsive self-mutilation** (lip/finger biting — pathognomonic!)

🔗 WHY NEUROLOGICAL FEATURES?

Brain depends almost entirely on **purine salvage** (negligible de novo). HGPRT loss → brain cannot maintain nucleotide pool → dopaminergic neuron dysfunction → movement disorder + self-mutilation behaviour.

- ◆ **Deficiency:** Adenosine Deaminase (ADA)
- ◆ **Inheritance:** Autosomal recessive
- ◆ ~15% of all SCID cases

Biochemical Mechanism:

- ◆ ADA absent → deoxyadenosine → phosphorylated to **dATP** accumulates
- ◆ ↑↑ dATP → inhibits **ribonucleotide reductase**
- ◆ Ribonucleotide reductase inhibited → all dNTPs ↓ → DNA synthesis blocked
- ◆ Lymphocytes are most affected (highest ADA activity normally)

Clinical Features:

- ◆ ↓↓ T cells + B cells + NK cells = "**Bubble boy**" disease
- ◆ Recurrent severe infections from birth
- ◆ Failure to thrive

Treatment:

- ◆ PEG-ADA (enzyme replacement)
- ◆ Bone marrow transplant
- ◆ **Gene therapy** — first disease treated by gene therapy (1990, Ashanti DeSilva)

FEATURE	LESCH-NYHAN	ADA DEFICIENCY
Deficient enzyme	HGPRT	ADA (Adenosine Deaminase)
Accumulated metabolite	Uric acid + PRPP	dATP (deoxyadenosine triphosphate)
Primary affected cells	Neurons (basal ganglia), kidney	T and B lymphocytes
Key clinical sign	Self-mutilation + gout	Severe combined immunodeficiency
Inheritance	X-linked recessive	Autosomal recessive
Treatment	Allopurinol (for gout) — no neurological cure	Gene therapy / PEG-ADA / BMT

08 Nucleotide Analogs as Drugs

⚡ Synthetic nucleotide analogs exploit the machinery of nucleotide metabolism to **kill rapidly dividing cells** (anticancer) or **inhibit viral replication** (antiviral). Several are suicide inhibitors.

DRUG	ANALOG OF	TARGET ENZYME	MECHANISM	USE
6-Mercaptopurine (6-MP) Anticancer	Hypoxanthine	HGPRT (converts it), then inhibits de novo + salvage	Converted to 6-TIMP → inhibits PRPP amidotransferase → ↓ de novo purine synthesis. Also incorporated into DNA/RNA	Leukaemia (ALL, AML)
6-Thioguanine Anticancer	Guanine	HGPRT converts it; then multiple targets	Incorporated into DNA → strand breaks, mismatch repair activation → apoptosis	Leukaemia
5-Fluorouracil (5-FU) Anticancer	Uracil	Thymidylate synthase (TS) — suicide inhibitor	5-FdUMP + methylene-THF forms covalent ternary complex with TS → ↓ dTMP → ↓ DNA synthesis → "thymineless death"	Colon, breast, head-neck Ca
Methotrexate (MTX) Anticancer	Folate (not nucleotide)	Dihydrofolate Reductase (DHFR)	Inhibits DHFR → ↓ THF → ↓ one-carbon transfer → ↓ dTMP + ↓ purine synthesis (both). "Indirect" nucleotide analog	Leukaemia, lymphoma, autoimmune diseases, ectopic pregnancy
AZT (Zidovudine) Antiviral	Thymidine	HIV Reverse Transcriptase	Lacks 3'-OH → chain terminator (no next nucleotide can attach). Must be phosphorylated by cellular kinases first	HIV/AIDS
Acyclovir Antiviral	Guanosine (acyclic)	Viral thymidine kinase (activates it) → viral DNA polymerase	Selectively phosphorylated by viral TK (1000× better than cellular TK) → acyclo-GTP → chain terminator of viral DNA pol	Herpes simplex, VZV
Cytarabine (Ara-C) Anticancer	Cytidine (arabinose sugar)	DNA polymerase	Incorporated into DNA → blocks elongation. Also inhibits DNA polymerase directly	AML, CLL
Ciprofloxacin Antibiotic	Not nucleotide per se	Bacterial DNA Gyrase (Topoisomerase II) and Topoisomerase IV	Stabilises enzyme-DNA cleavage complex → double-strand breaks → bacterial cell death. No direct structural analog of nucleotide	Broad-spectrum bacteria
Mycophenolate & Ribavirin Immunosuppressant	Purine precursors	IMP Dehydrogenase	Inhibits conversion of IMP to Xanthosine-5-P → ↓ GMP synthesis in rapidly dividing B/T cells	Organ transplant, Hepatitis C
Hydroxyurea Anticancer	-	Ribonucleotide Reductase (RNR)	Inhibits conversion of ribonucleotides to deoxyribonucleotides → ↓ DNA synthesis	Melanoma, Sickle Cell (↑ HbF)

🔗 WHY 5-FU IS A SUICIDE INHIBITOR

5-FdUMP (active metabolite) forms a **covalent ternary complex** with thymidylate synthase AND N⁵,N¹⁰-methylene-THF. The enzyme is irreversibly inactivated. Because it uses the enzyme's own mechanism to trap it, this is the definition of **mechanism-based (suicide) inhibition**. Folate co-factor is simultaneously depleted.

⚠️ **EQ Classic:** "Ciprofloxacin is an antibiotic but 6-MP is an anticancer drug [2023]" — They both affect DNA/nucleotide metabolism but via completely different mechanisms and targets (CIP → bacterial gyrase vs 6-MP → de novo purine synthesis + DNA incorporation).

▶ TWO PATHWAYS USING CARBAMOYL PHOSPHATE – THE KEY DISTINCTION

Mitochondria
 $\text{NH}_3 + \text{CO}_2 + 2\text{ATP}$
↓ **CPS I** (activated by N-AcGlu)

Carbamoyl-P (mitochondrial)

↓ OTC + Aspartate

Citrulline → Urea Cycle

Defect → Hyperammonaemia

SEPARATE
COMPARTMENTS**Cytosol**
 $\text{Glutamine} + \text{CO}_2 + 2\text{ATP}$
↓ **CPS II** (inhibited by UMP)

Carbamoyl-P (cytosolic)

↓ + Aspartate (ATCase)

Carbamoyl Aspartate → Pyrimidine ring

→ Orotate → UMP → CTP

Defect → Orotic Aciduria

FEATURE	OROTIC ACIDURIA TYPE I (UMP SYNTHASE DEFICIENCY)	OTC DEFICIENCY (UREA CYCLE)
Block location	OPRT + ODCase (UMP synthase — bi-functional)	Ornithine transcarbamoylase (mitochondria)
Orotic acid ↑	Yes ↑↑	Yes ↑
Ammonia ↑	No (urea cycle intact)	Yes ↑↑ (main feature)
Megaloblastic anaemia	Yes (cannot make pyrimidines)	No
Treatment	Uridine (bypasses block + inhibits CPS II)	Protein restriction + sodium benzoate; liver transplant

 PURINE METABOLISM VIVAS
What are the sources of N and C atoms in the purine ring? Draw and label.

PYQ 2013, 2015-S

The purine ring has 9 atoms: 4C + 5N (actually 5C + 4N in the ring). The donors:

- **Glycine:** contributes C4, C5, N7 (an entire glycine molecule is incorporated)
- **Glutamine:** contributes N3, N9 (amide nitrogen); also N at C2 and C6
- **Aspartate:** contributes N1 (via adenylosuccinate)
- **N¹⁰-Formyl-THF:** contributes C2 and C8 (one-carbon folate units)
- **CO₂:** contributes C6

Mnemonic: "GAGA + CO₂" → **G**lycine, **A**spartate, **G**lutamine×2 (different positions), **f**ormyl-THF (C2,C8), **C**O₂ (C6)

The first committed step uses PRPP as the ribose scaffold. The ring is assembled on the sugar — 10 enzymatic steps to form IMP.

What is the salvage pathway? What is its clinical significance?

PYQ 2018, 2025

Salvage pathway: Recycles pre-formed purine bases (from nucleic acid degradation or diet) back to nucleotides, using PRPP as the phosphoribosyl donor.

- HGPRT: Hypoxanthine + PRPP → IMP; Guanine + PRPP → GMP
- APRT: Adenine + PRPP → AMP

Clinical significance:

- Energetically cheap (no de novo 10-step pathway needed)
- **Brain survival:** brain has minimal de novo capacity → depends entirely on salvage
- **HGPRT deficiency** → Lesch-Nyhan syndrome (gout + neurological + self-mutilation)
- Drug exploitation: 6-MP and 6-TG are converted by HGPRT to active forms (anti-leukemia drugs work THROUGH salvage)
- Provides feedback regulation: salvage products (IMP, GMP, AMP) inhibit de novo synthesis

How does allopurinol lower uric acid? Why is it called a suicide inhibitor?

PYQ 2012, 2019

Allopurinol (isomer of hypoxanthine with C7-N8 transposed) is acted upon by **xanthine oxidase**:

1. XO oxidises allopurinol → **Oxipurinol (alloxanthine)**
2. Oxipurinol binds tightly to the **reduced (Mo⁴⁺) form** of XO → forms long-lived inhibitory complex
3. The enzyme's own catalytic mechanism produces the inhibitor → **mechanism-based / suicide inhibition**

Results:

- ↓ conversion of hypoxanthine and xanthine to uric acid
- Hypoxanthine + xanthine accumulate (10× more soluble than urate → safely excreted)
- PRPP accumulates → but overall PRPP is consumed → de novo synthesis also decreases
- Reduces serum and urinary uric acid by ~60–70%

Differentiate between purine de novo synthesis and pyrimidine de novo synthesis.

Core Comparison

FEATURE	PURINE	PYRIMIDINE
Ring built	ON ribose-5-P (preformed sugar scaffold)	FREE ring first, THEN attached to PRPP
Rate-limiting step	Glutamine:PRPP amidotransferase	CPS II
Common intermediate	IMP → AMP or GMP	UMP → UDP → UTP → CTP or dTMP
Atom donors	5 different donors (Gly, Gln, Asp, THF, CO ₂)	3 donors (Asp, CO ₂ , Gln)
End product of catabolism	Uric acid (humans)	CO ₂ + NH ₃ + β-alanine/β-AIB
Key drug target	PRPP amidotransferase (6-MP), HGPRT, XO (allopurinol)	Thymidylate synthase (5-FU), DHFR (MTX), CPS II

Why does ADA deficiency cause immunodeficiency specifically (not other systems)?

PYQ 2011-S

The selective lymphocyte toxicity of ADA deficiency is due to **unique characteristics of lymphocytes**:

- Lymphocytes have **high levels of deoxynucleoside kinase** (TdK) relative to 5'-nucleotidase → deoxyadenosine is preferentially phosphorylated to dAMP → dADP → **dATP**
- Other cell types have high 5'-nucleotidase activity → dephosphorylate dAMP back to adenosine (before toxic accumulation)
- Lymphocytes also have **high mitochondrial deoxyguanosine kinase** sensitivity
- dATP specifically inhibits ribonucleotide reductase → all dNTPs ↓ → S-phase DNA synthesis block → apoptosis
- Additionally, dATP causes direct DNA strand breaks and activates apoptotic pathways

Other tissues: High 5'-nucleotidase activity rapidly dephosphorylates dAMP → no toxic accumulation → spared.

What is orotic aciduria? How do you distinguish it from OTC deficiency?

Advance Topic

Both conditions → ↑ orotic acid in urine. Key differentiating features:

- OTC deficiency:** Urea cycle block → carbamoyl-P accumulates in mitochondria → overflows into cytosol → pyrimidine synthesis goes into overdrive → orotate ↑. BUT: blood ammonia ↑↑↑ (main feature). No megaloblastic anaemia (UMP synthase intact can still process orotate).
- Orotic Aciduria Type I:** UMP synthase deficiency in cytosol → orotate cannot be converted to UMP → orotate ↑. Ammonia NORMAL (urea cycle intact). **Megaloblastic anaemia** present (cannot make pyrimidines → ↓ DNA synthesis in RBC precursors).

Treatment difference: Orotic aciduria → uridine supplements. OTC deficiency → protein restriction + ammonia scavengers + liver transplant.

Allopurinol lowers the uric acid concentration of blood.

PYQ 2012-S, 2019

Standard Exam Answer Draft:

1. **Target:** Chronic Gout is caused by hyperuricaemia, resulting in Monosodium Urate crystal precipitation in joints.
2. **Therapeutic Goal:** Lower serum uric acid below 6.0 mg/dL to dissolve existing crystals and prevent new deposits.
3. **Mechanism:** Allopurinol is a structural isomer of hypoxanthine. Xanthine Oxidase (XO) converts it to Oxipurinol (Alloxanthine).
4. **Inhibition Class:** Oxipurinol binds tightly and irreversibly to the Mo^{4+} active site of XO, acting as a **suicide inhibitor**.
5. **Metabolic Shift:** Conversion of hypoxanthine and xanthine to uric acid is blocked. Soluble hypoxanthine and xanthine accumulate and are excreted, while serum uric acid levels drop.
6. **De Novo Downregulation:** Clearance of PRPP by salvage pathways reduces substrates for de novo purine synthesis, further lowering urate levels.

Gout is precipitated by alcohol intake.

PYQ 2012-S, 2016-S, 2018-S, 2021

Standard Exam Answer Draft:

1. **Metabolic Block:** Oxidation of ethanol by Alcohol Dehydrogenase generates NADH. High NADH/NAD⁺ ratio favors the reduction of pyruvate to lactate.
2. **Excretion Competition:** Lactate accumulates and competitively inhibits the organic anion transporter (URAT1) in renal proximal tubule cells, preventing renal uric acid secretion and elevating serum urate.
3. **Purine Load:** Yeast in fermented drinks (such as beer) contains high levels of guanosine, which is catabolized directly into uric acid.
4. **ATP Turnover:** High acetate production in liver cells drives ATP conversion to AMP. Accumulation of AMP feeds into the purine catabolic pathway, accelerating urate synthesis.

Adenosine deaminase deficiency leads to immunodeficiency state.

PYQ 2011-S

Standard Exam Answer Draft:

1. **Metabolic Defect:** Absence of Adenosine Deaminase prevents the conversion of adenosine to inosine, and deoxyadenosine to deoxyinosine.
2. **Metabolite Accumulation:** Deoxyadenosine accumulates and is phosphorylated by lymphoid deoxycytidine kinase to form toxic **dATP**.
3. **Enzyme Inhibition:** High dATP binds to the activity site of **Ribonucleotide Reductase (RNR)**, shutting down the conversion of all ribonucleotides to deoxyribonucleotides.
4. **DNA Synthesis Arrest:** DNA replication is arrested in T, B, and NK cells, inducing apoptosis. This results in the profound lymphopenia characteristic of Severe Combined Immunodeficiency (SCID).

Rate-limiting step of purine de novo

Glutamine:PRPP amidotransferase (first committed step)

Rate-limiting step of pyrimidine de novo

CPS II (cytosolic carbamoyl phosphate synthetase)

Common intermediate — purines

IMP (Inosine Monophosphate)

Common intermediate — pyrimidines

UMP (Uridine Monophosphate)

HGPRT deficiency → ?

Lesch-Nyhan syndrome (X-linked, gout + self-mutilation)

ADA deficiency → ?

SCID (dATP accumulates → RNR inhibition → lymphocyte death)

Allopurinol mechanism type

Suicide/mechanism-based inhibitor (via oxipurinol)

5-FU target enzyme

Thymidylate synthase (inhibits dTMP synthesis)

Methotrexate target enzyme

DHFR (Dihydrofolate reductase)

AZT mechanism

Chain terminator of HIV reverse transcriptase (no 3'-OH)

End product of purine catabolism in humans

Uric acid (uricase gene silenced)

Orotic aciduria + anaemia → ?

UMP synthase deficiency. Treat with Uridine.

Orotic aciduria + hyperammonaemia → ?

OTC deficiency (urea cycle, NOT pyrimidine pathway)

Gout + alcohol biochemical link

Lactate competes with urate for renal secretion + ATP → AMP → uric acid

6-MP + allopurinol interaction

XO normally degrades 6-MP; allopurinol blocks XO → 6-MP toxic; reduce dose to 25%

MODULE 05 MASTER MNEMONICS

● PURINE RING DONORS

"Gar, he makes funny coloured xylophones"

G=Gly(C4,C5,N7), He=Gln(N3,N9),
Makes=Asp(N1), F=Formyl-THF(C2,C8),
C=CO₂(C6)

● PYRIMIDINE DONORS - SIMPLE

"Aspartate + Glutamine + CO₂ = ACG builds pyrimidines"

● PURINES VS PYRIMIDINES IN DNA/RNA

PURe As Gold = Purines: A, G
CUT the Py = Pyrimidines: C, U(RNA),
T(DNA)

● LESCH-NYHAN TRIAD

"GUN" = Gout, Uric acid↑, Neurological (self-mutilation)

WBUHS Biochemistry - Module 05: Nucleotide Chemistry & Metabolism

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